Worked Examples · 例题

Circular Motion & Gravitational Fields

Translational motion			Circular motion		
Quantity	Unit	Relationships	Quantity	Unit	Relationships
Displacement (s)	m		Angular displacement (θ)	rad	
Speed (v)	$m\,s^{-1}$	$v = \frac{\Delta s}{\Delta t}$	Angular speed (θ)	$rad\,s^{-1}$	$\omega = \frac{\Delta \theta}{\Delta t}$

Look at Figure 14.3.

$$\omega = \frac{\Delta \theta}{\Delta t}$$

but:

$$\Delta \theta = \frac{AB}{r}$$

therefore:

$$\omega = \frac{AB}{r \Delta t}$$

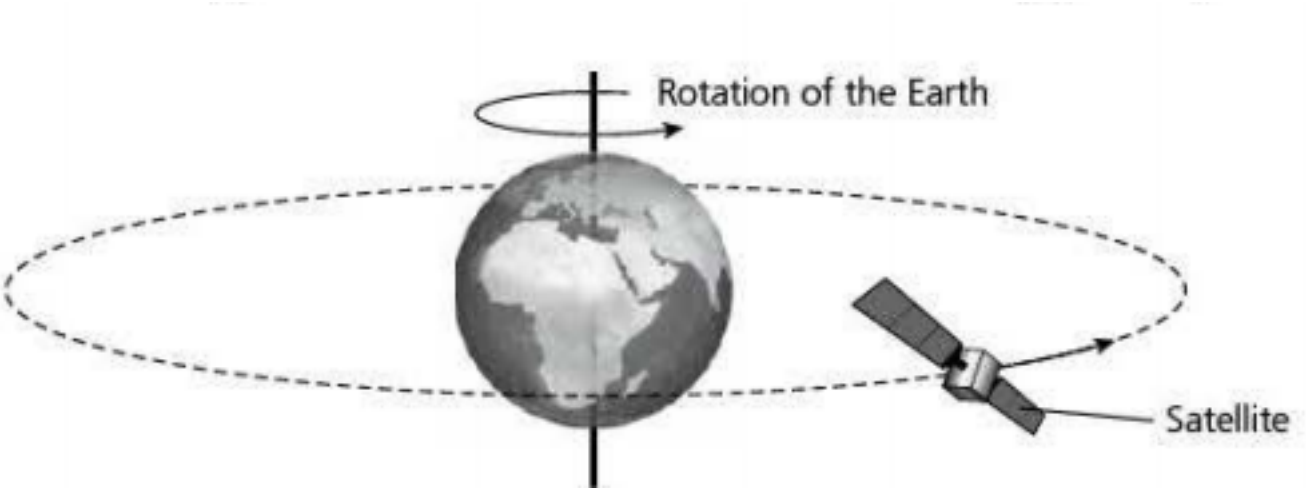
$$\frac{AB}{\Delta t} = \frac{\text{distance travelled}}{\text{time taken}} = v$$

$$\text{so } \omega = \frac{v}{r}$$

or, rearranging the formula:

$$\boldsymbol{v = \omega r}$$

circgrav_00.png



circgrav_04.png

Worked example

A satellite is to be placed in a polar orbit 100 km above the Earth's surface.

Calculate:

(a) the period of the orbit

(b) the speed of the satellite

(mass of Earth = 6.0×10^{24} kg, radius of Earth = 6.4×10^{3} km)

Answer

(a) orbital radius of the satellite = Earth's radius + height of the satellite above the surface

$= (6.4 \times 10^{3} + 100) = 6.5 \times 10^{3}\,\text{km} = 6.5 \times 10^{6}\,\text{m}$

$$\omega = \sqrt{\frac{GM}{r^3}} = \sqrt{\frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24}}{(6.5 \times 10^{6})^3}} = 1.25 \times 10^{-3}\,\text{rad s}^{-1}$$

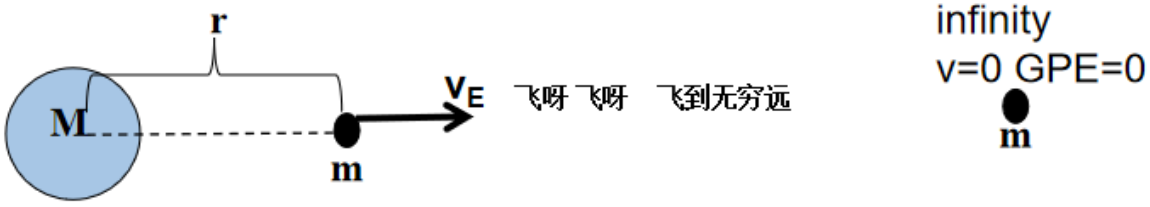
$$T = \frac{2\pi}{\omega} = \frac{2\pi}{1.25 \times 10^{-3}} = 5.0 \times 10^{3}\,\text{s} = 1.4\,\text{h}$$

(b) $v = \omega r$

orbital radius = 6.5×10^{3} km

$v = 1.2 \times 10^{-3} \times 6.5 \times 10^{3} = 7.8\,\text{km s}^{-1}$

circgrav_05.png


r
M
m
v_E
飞呀 飞呀 飞到无穷远
infinity
v=0 GPE=0
m

If a body is fired from the Earth's surface with sufficient speed, it can escape from the Earth's gravitational field.

(a) Calculate the potential at the Earth's surface.

(b) State and explain how much energy a body of unit mass would need to be given to escape from the Earth's field.

(c) Calculate the minimum speed at which the body must be fired to escape.

Answer

(a) $\phi = -\frac{GM}{r} = -\frac{-6.67 \times 10^{-11} \times 6.0 \times 10^{24}}{6.4 \times 10^{6}} = -6.25 \times 10^{7}\,\text{J kg}^{-1}$

(b) 6.25×10^{7} J, the energy required to reach infinity, zero potential energy.

(c) $E_k = \frac{1}{2}mv^2$, which leads to:

$$v = \sqrt{\frac{2E_k}{m}} = \sqrt{\frac{2 \times 6.25 \times 10^{7}}{1}} = 1.1 \times 10^{4}\,\text{m s}^{-1}$$

circgrav_07.png

Worked example

Calculate the height above the Earth that a satellite must be placed for it to orbit in a geostationary manner.

(mass of Earth = 6.0×10^{24} kg, radius of Earth = 6.4×10^{6} m)

Answer

Time period required for a geostationary orbit is 24 h = 86 400 s

$$\omega = \frac{2\pi}{T}$$

$$\omega = \sqrt{\frac{GM}{r^3}}$$

$$\text{so } \frac{2\pi}{T} = \sqrt{\frac{GM}{r^3}}$$

and

$$r^3 = \frac{GMT^2}{(2\pi)^2} = \frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 86400^2}{4\pi^2} = 7.57 \times 10^{22}$$

$$r = \sqrt[3]{7.57 \times 10^{22}} = 4.23 \times 10^{7}$$

This is the radius of the satellite's orbit. The radius of the Earth is 6.4×10^{6} m, so the height of the satellite above the Earth's surface is:

$$42.3 \times 10^{6} - 6.4 \times 10^{6} = 3.59 \times 10^{7}\ \text{m} \approx 3.6 \times 10^{7}\ \text{m}$$

circgrav_09.png

A spherical planet has a mass *M* and radius *R*. The planet may be considered to have all its mass concentrated at its centre.

A rocket, of mass *m* is launched from the surface of the planet.

The rocket engines are stopped when the rocket is at a height *R* above the surface of the planet, as shown in Fig 1.1.

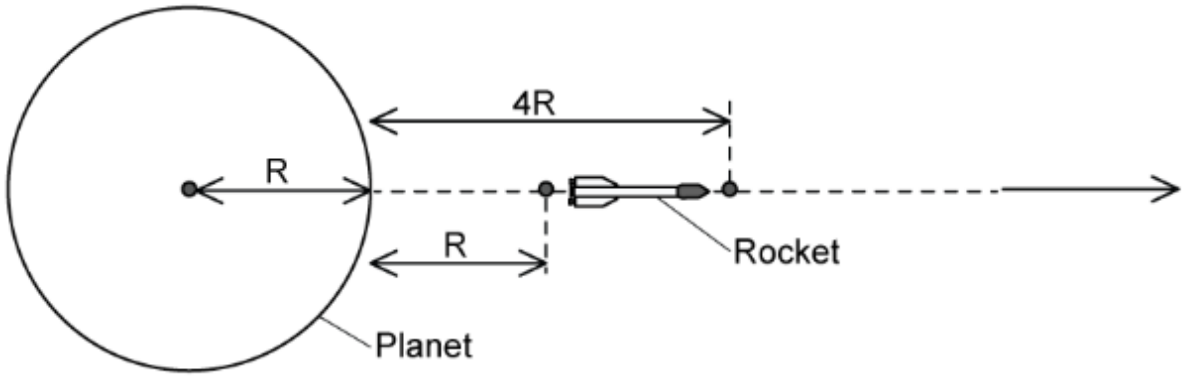


Fig. 1.1

The change in gravitational potential energy of the rocket as it travels from a height *R* to a height 4*R* above the planet's surface is equal to

circgrav_11.png

Write expressions for the initial and final gravitational potential energy:

- $E_p = -\frac{GMm}{r}$
- $Initial\ GPE = -\frac{GMm}{2R}$

  AND
- $Final\ GPE = -\frac{GMm}{5R}$ [1 mark]

Consider the change in gravitational potential energy of the rocket:

- $\Delta E_p = Final\ GPE - Initial\ GPE$
- $\Delta E_p = \left(-\frac{GMm}{5R}\right) - \left(-\frac{GMm}{2R}\right)$ [1 mark]
- $\Delta E_p = \left(\frac{GMm}{2R}\right) - \left(\frac{GMm}{5R}\right) \quad \Rightarrow \quad \left(\frac{5GMm}{10R}\right) - \left(\frac{2GMm}{10R}\right)$
- $\Delta E_p = \frac{3GMm}{10R}$
- Hence $x = \frac{3}{10}$ [1 mark]

At a distance R away from the surface of the planet the rocket has a velocity $2v$ and at a distance of $4R$ away from the planet a velocity v.

Show that the expression for the thermal energy lost ΔQ to the atmosphere of the planet as the rocket moves from R to $4R$ is given by

$$\Delta Q = \frac{3m}{2}\left(v^2 - \frac{GM}{5R}\right)$$

Obtain an expression for the change in kinetic energy of the rocket:

- $\Delta KE = Final\ KE - Initial\ KE$
- $\Delta KE = \frac{1}{2}mv^2 - \frac{1}{2}m(2v)^2 = mv^2\left(\frac{1}{2} - \frac{4}{2}\right)$
- $\Delta KE = -\frac{3}{2}mv^2$ [1 mark]

Consider the conservation of energy:

- $\Delta E_p + \Delta KE + \Delta Q = 0$ [1 mark]

Substitute the known expressions and rearrange to make ΔQ the subject:

- Change in GPE: $\Delta E_p = \frac{3GMm}{10R}$ (from **(a)**)
- $\Delta Q = -\Delta E_p - \Delta KE$
- $\Delta Q = -\frac{3GMm}{10R} - \left(-\frac{3mv^2}{2}\right)$ [1 mark]
- $\Delta Q = \frac{3mv^2}{2} - \frac{3GMm}{10R}$
- $\Delta Q = \frac{3m}{2}\left(v^2 - \frac{GM}{5R}\right)$ [1 mark]

The rocket travels towards a moon orbiting the planet

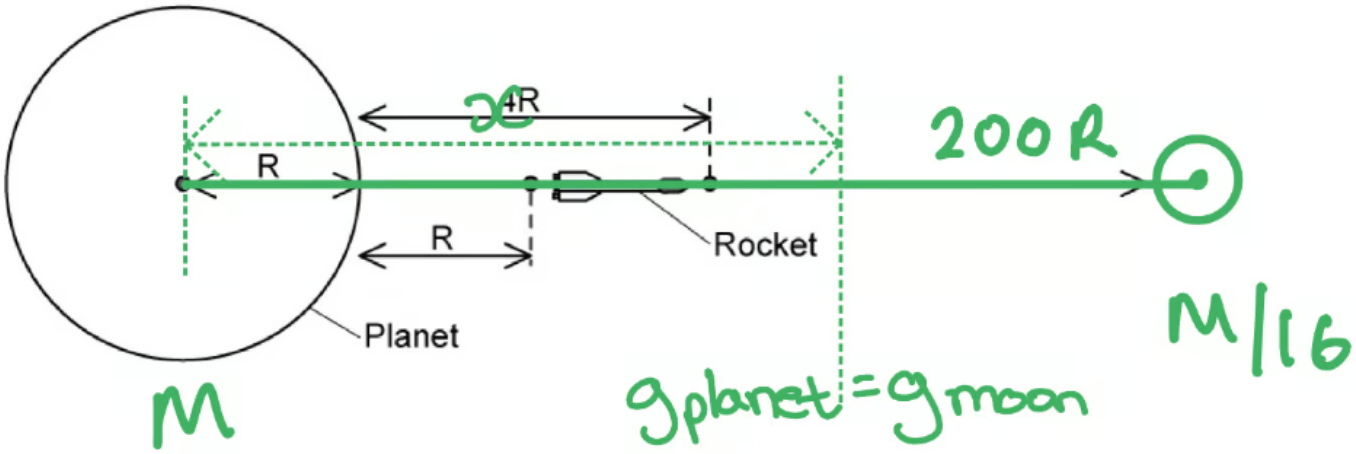
The distance from the centre of the planet to the centre of the moon is 200R. The moon has a mass $\frac{M}{16}$ where M is the mass of the planet.

Determine the distance, in terms of R, from the centre of the planet at which

$$\frac{\Delta E_P}{\Delta r} = 0$$

Give your answer to the nearest whole number.

circgrav_18.png


4R
R
R
Rocket
Planet
200R
M/16
M
gplanet = gmoon

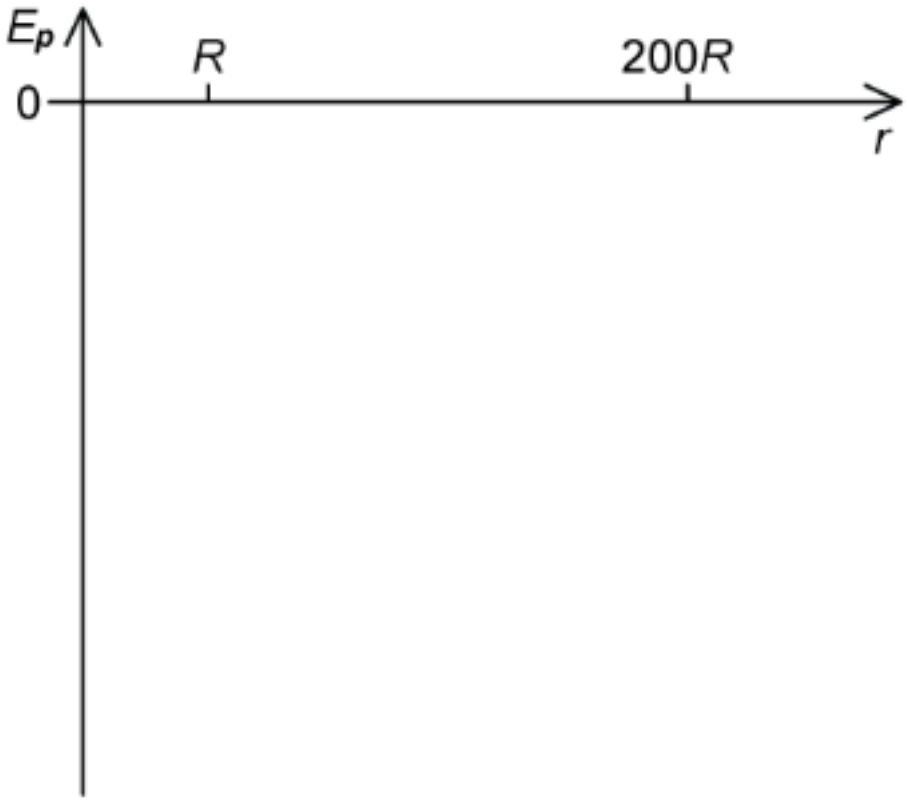
- Gravitational field strength is equal to the gradient of the potential-distance graph i.e. $g = \frac{\Delta E_P}{\Delta r}$
- Hence, the point where $\frac{\Delta E_P}{\Delta r} = 0$ is where $g_{planet} = -g_{moon}$
- Field strength due to planet: $g_P = \frac{GM}{x^2}$ [1 mark]
- Field strength due to moon: $g_M = \frac{G\left(\frac{M}{16}\right)}{(200R - x)^2}$ [1 mark]

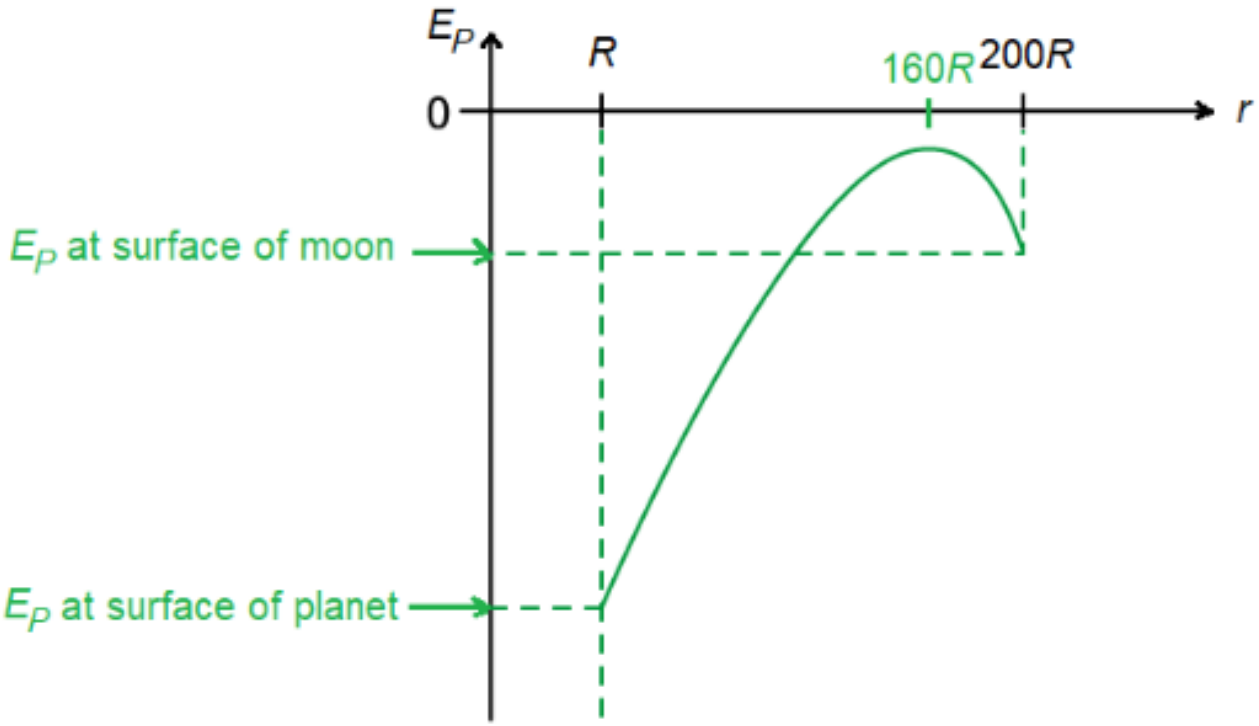
Equate the expressions for gravitational field strength to obtain a value for x:

circgrav_20.png

Equate the expressions for gravitational field strength to obtain a value for x:

- $g_P = -g_M$
- $\frac{GM}{x^2} = -\frac{GM}{16(200R - x)^2}$ [1 mark]
- $x^2 = 16(200R - x)^2 \quad \Rightarrow \quad \sqrt{x^2} = \sqrt{16(200R - x)^2}$
- $x = 4(200R - x) = 800R - 4x$
- $x = \frac{800}{5}R = 160R$ [1 mark]


E_p
0
R
200R
r



circgrav_24.png

(a) (i) Define gravitational potential at a point.

Gravitational potential is defined as the work done per unit mass in moving that mass from infinity to the point.

..... [2]

(ii) Starting from the equation for the gravitational potential due to a point mass, show that the gravitational potential energy E_P of a point mass m at a distance r from another point mass M is given by

$$E_P = -\frac{GMm}{r}$$

where G is the gravitational constant.

potential = $-GM / r$

$\therefore E_P$ = potential $\times$ m = $(-GM / r) \times m = -GMm / r$.

[1]

(b) Fig. 1.1 shows the path of a comet of mass 2.20×10^{14} kg as it passes around a star of mass 1.99×10^{30} kg.

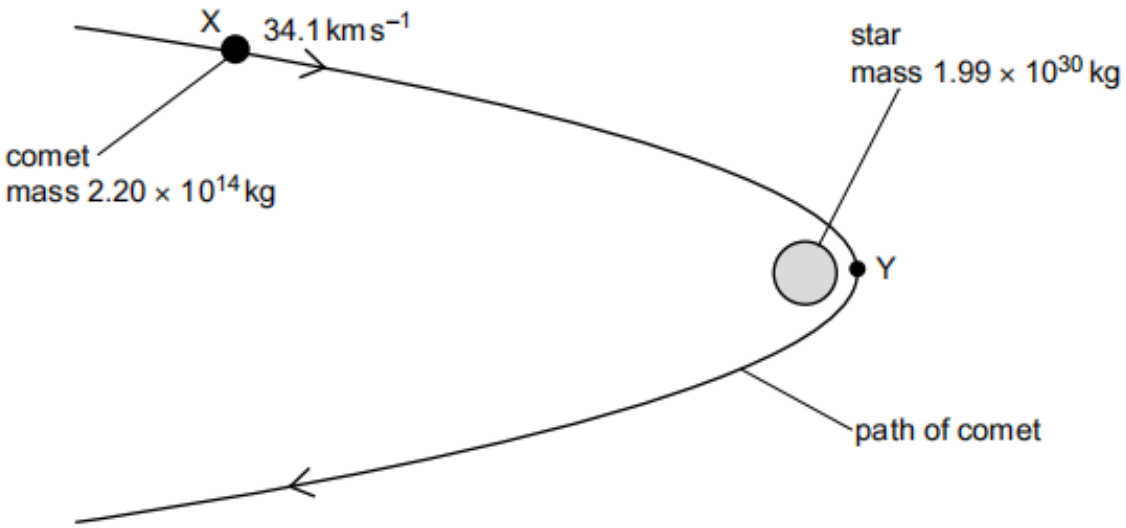


Fig. 1.1 (not to scale)

At point X, the comet is 8.44×10^{11} m from the centre of the star and is moving at a speed of 34.1 km s^{-1}.

At point Y, the comet passes its point of closest approach to the star. At this point, the comet is a distance of 6.38×10^{10} m from the centre of the star.

Both the comet and the star can be considered as point masses at their centres.

(i) Calculate the magnitude of the change in the gravitational potential energy ΔE_P of the comet as it moves from position X to position Y.

$\Delta E_P = 6.67 \times 10^{-11} \times 1.99 \times 10^{30} \times 2.20 \times 10^{14} \times [1/(6.38 \times 10^{10}) - 1/(8.44 \times 10^{11})]$

$\Delta E_P =$ 4.23×10^{23} J [2]

(ii) State, with a reason, whether the change in gravitational potential energy in **(b)(i)** is an increase or a decrease.

The gravitational force acting on the comet moves it closer to the star, increasing the kinetic energy of the comet and therefore [1] decreasing its potential energy. The change in (b)(i) is therefore a decrease.

(iii) Use your answer in **(b)(i)** to determine the speed, in km s^{-1}, of the comet at point Y.

$\Delta E_P = \frac{1}{2}m(v_2^2 - v_1^2)$

$\therefore 4.23 \times 10^{23} = \frac{1}{2} \times 2.20 \times 10^{14} \times (v2 - 34100^2)$

$\therefore v = 70800 \text{ m s}^{-1} = 70.8 \text{ km s}^{-1}$

speed = ... 70.8 km ..... km s^{-1} [3]

(c) A second comet passes point X with the same speed as the comet in **(b)** and travelling in the same direction. This comet is gradually losing mass. The mass of this comet when it passes point X is the same as the mass of the comet in **(b)**.

Suggest, with a reason, how the path of the second comet compares with the path shown in Fig. 1.1.

The mass of the comet cancels out in the equations relating potential and kinetic energies. Therefore, the change in mass does not affect the path of the comet. [1]

[Total: 10]

12 Motion in a circle

12.1 Kinematics of uniform circular motion

Candidates should be able to:

1 define the radian and express angular displacement in radians

2 understand and use the concept of angular speed

3 recall and use $\omega = 2\pi / T$ and $v = r\omega$

12.2 Centripetal acceleration

Candidates should be able to:

1 understand that a force of constant magnitude that is always perpendicular to the direction of motion causes centripetal acceleration

2 understand that centripetal acceleration causes circular motion with a constant angular speed

3 recall and use $a = r\omega^2$ and $a = v^2 / r$

4 recall and use $F = mr\omega^2$ and $F = mv^2 / r$

13 Gravitational fields

13.1 Gravitational field

Candidates should be able to:

1 understand that a gravitational field is an example of a field of force and define gravitational field as force per unit mass

2 represent a gravitational field by means of field lines

13.2 Gravitational force between point masses

Candidates should be able to:

1 understand that, for a point outside a uniform sphere, the mass of the sphere may be considered to be a point mass at its centre

2 recall and use Newton's law of gravitation $F = Gm_1m_2/r^2$ for the force between two point masses

3 analyse circular orbits in gravitational fields by relating the gravitational force to the centripetal acceleration it causes

4 understand that a satellite in a geostationary orbit remains at the same point above the Earth's surface, with an orbital period of 24 hours, orbiting from west to east, directly above the Equator

13.3 Gravitational field of a point mass

Candidates should be able to:

1 derive, from Newton's law of gravitation and the definition of gravitational field, the equation $g = GM/r^2$ for the gravitational field strength due to a point mass

2 recall and use $g = GM/r^2$

3 understand why g is approximately constant for small changes in height near the Earth's surface

circgrav_31.png

13.4 Gravitational potential

Candidates should be able to:

1 define gravitational potential at a point as the work done per unit mass in bringing a small test mass from infinity to the point

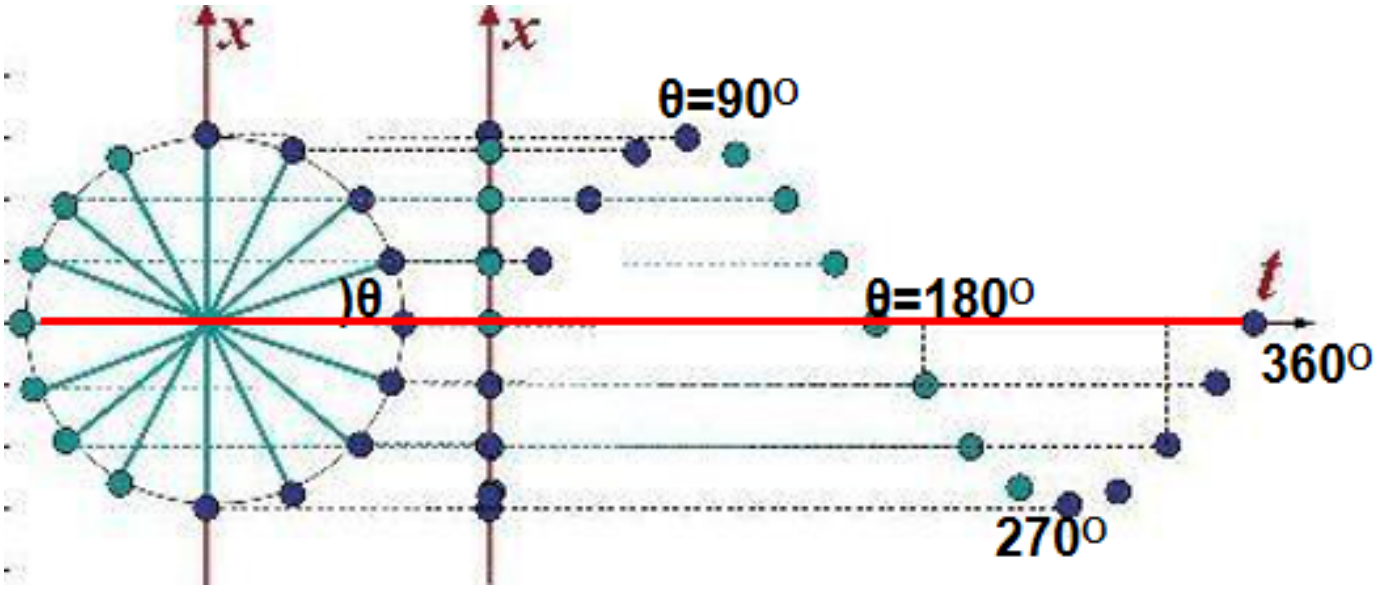
2 use $\phi = -GM/r$ for the gravitational potential in the field due to a point mass

3 understand how the concept of gravitational potential leads to the gravitational potential energy of two point masses and use $E_p = -GMm/r$

circgrav_32.png

Worked Examples · 例题

Oscillations



osc_01.png

Figure 18.10 shows displacement–time graphs for two identical oscillators. Calculate the phase difference between the two oscillations. Give your answer in degrees and in radians.

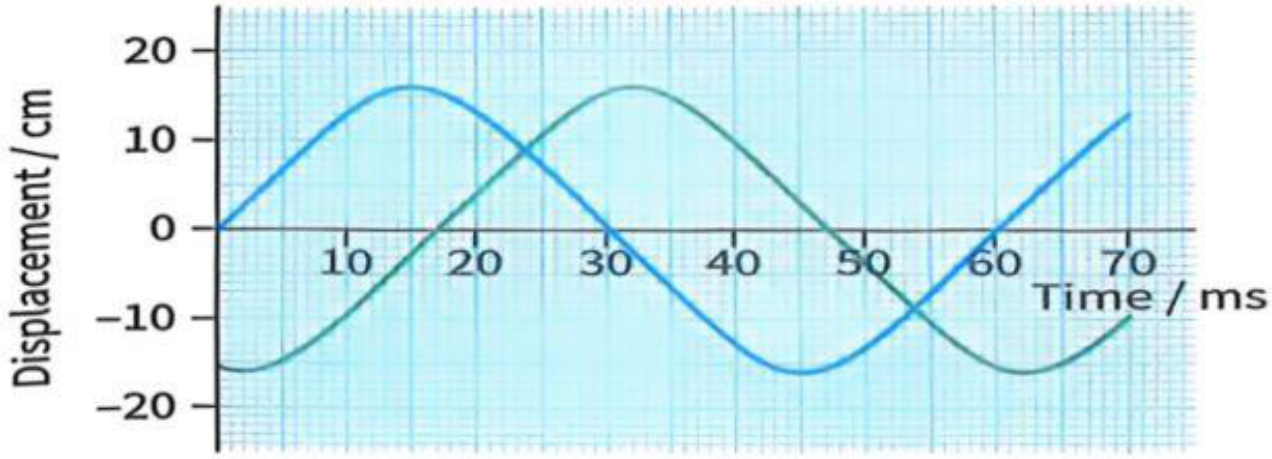


Figure 18.10: The displacement–time graphs of two oscillators with the same period.

Step 1 Measure the time interval t between two corresponding points on the graphs.

$t = 17\,\text{ms}$

Step 2 Determine the period T for one complete oscillation.

$T = 60\,\text{ms}$

Hint: *Remember that a complete oscillation is when the object goes from one side to the other and back again.*

osc_03.png

Step 3 Now you can calculate the phase difference as a fraction of an oscillation.

phase difference = fraction of an oscillation

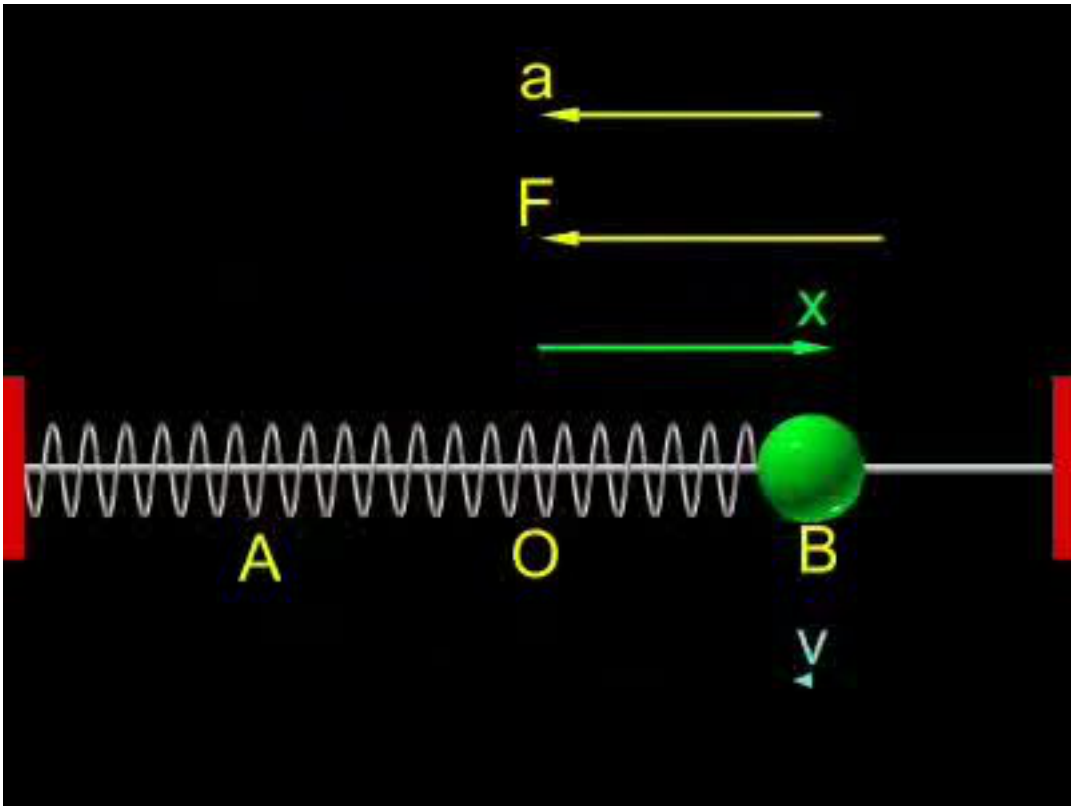
Therefore:

$$\text{phase difference} = \frac{t}{T}$$
$$= \frac{17}{60}$$
$$= 0.283 \text{ oscillations}$$

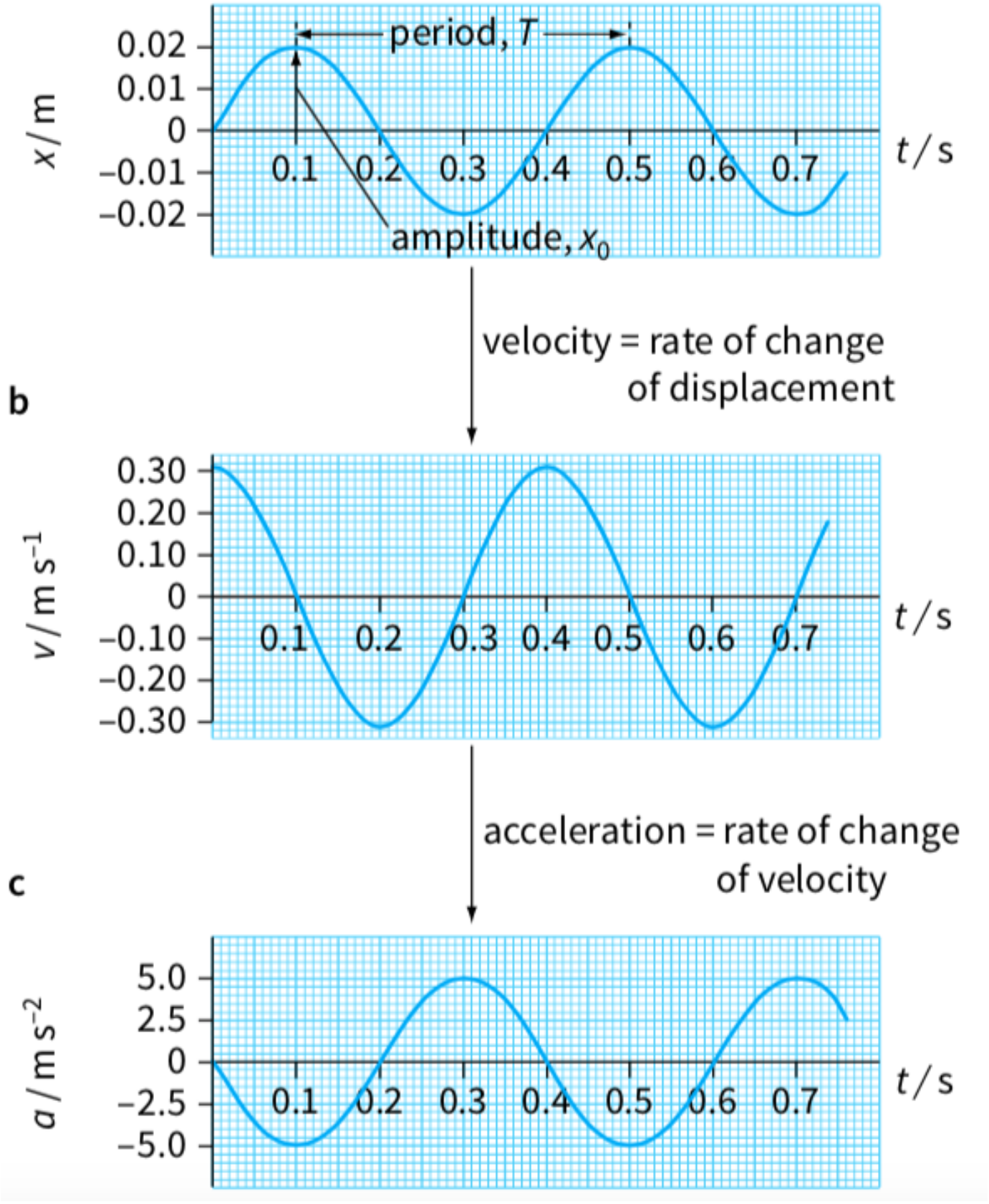
Step 4 Convert to degrees and radians. There are 360° and 2π rad in one oscillation.

$$\text{phase difference} = 0.283 \times 360°$$
$$= 102° \approx 100°$$
$$\text{phase difference} = 0.283 \times 2\pi \text{ rad}$$
$$= 1.78 \text{ rad} \approx 1.8 \text{ rad}$$

osc_04.png



osc_06.png


period, T
0.02
0.01
0
−0.01
−0.02
x / m
0.1
0.2
0.3
0.4
0.5
0.6
0.7
t / s
amplitude, x_0
velocity = rate of change of displacement
b
0.30
0.20
0.10
0
−0.10
−0.20
−0.30
v / m s^{-1}
0.1
0.2
0.3
0.4
0.5
0.6
0.7
t / s
acceleration = rate of change of velocity
c
5.0
2.5
0
−2.5
−5.0
a / m s^{-2}
0.1
0.2
0.3
0.4
0.5
0.6
0.7
t / s

osc_07.png

A pendulum oscillates with frequency 1.5 Hz and amplitude 0.10 m. If it is passing through its equilibrium position when $t = 0$, write an equation to represent its displacement x in terms of amplitude x_0, angular frequency ω and time t. Determine its displacement when $t = 0.50$ s.

Step 1 Select the correct equation. In this case, the displacement is zero when $t = 0$, so we use the sine form:

$$x = x_0 \sin \omega t$$

Step 2 From the frequency f, calculate the angular frequency ω:

$$\begin{aligned} \omega &= 2\pi f \\ &= 2 \times \pi \times 1.5 \\ &= 3.0\pi \end{aligned}$$

Step 3 Substitute values in the equation: $x_0 = 0.10$ m, so:

$$x = 0.10 \sin (3.0\pi t)$$

Hint: *Remember to put your calculator into radian mode.*

Step 4 To find x when $t = 0.50$ s, substitute for t and calculate the answer:

$$\begin{aligned} x &= 0.10 \sin(2\pi \times 1.5 \times 0.50) \\ &= 0.10 \sin(4.713) \\ &= -0.10 \text{ m} \end{aligned}$$

This means that the pendulum is at the extreme end of its oscillation; the minus sign means that it is at the negative or left-hand end, assuming you have chosen to consider displacements to the right as positive.

(If your calculation went like this:

$$\begin{aligned} x &= 0.10 \sin(2\pi \times 1.5 \times 0.50) \\ &= 0.10 \sin(4.713) \\ &= -8.2 \times 10^{-3} \text{ m} \end{aligned}$$

then your calculator was incorrectly set to work in degrees, not radians.)

osc_08.png

The pendulum of a grandfather clock swings from one side to the other in 1.00 s. The amplitude of the oscillation is 12 cm.

a Calculate:

i the period of its motion

ii the frequency

iii the angular frequency.

b Write an equation of the form $a = -\omega^2 x$ to show how the acceleration of the pendulum bob depends on its displacement.

c Calculate the maximum speed of the pendulum bob.

d Calculate the speed of the bob when its displacement is 6 cm.

osc_09.png

a **i** 2.0 s

ii frequency $= 1/T = 1/2.0 = 0.5$ Hz

iii angular frequency $\omega = 2\pi f = \pi$ rad s^{-1} or 3.14 rad s^{-1}

b $a = -\omega^2 x = -9.87x$ or $a = -\pi^2 x$

c maximum speed $= \omega x_0 = 3.14 \times 12 =$ 37.6 cm s^{-1} $\approx$ 38 cm s^{-1}

d speed at $x = 6$ cm is

$$v = \omega\sqrt{(x_0^2 - x^2)} = \omega\sqrt{(144 - 36)}$$

$$= 32.6\,\text{cm s}^{-1} \approx 33\,\text{cm s}^{-1}$$

Figure 18.24 shows how the velocity v of a 2.0 kg mass was found to vary with time t during an investigation of the s.h.m. of a pendulum. Use the graph to estimate the following for the mass:

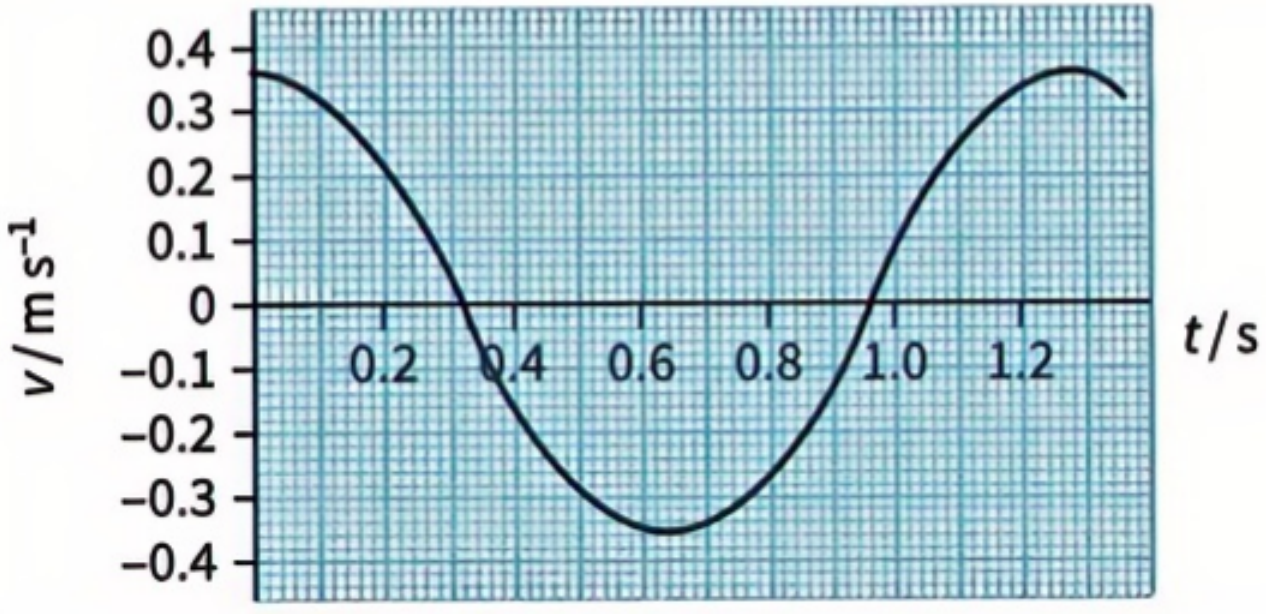
a its maximum velocity

b its maximum kinetic energy

c its maximum potential energy

d its maximum acceleration

e the maximum restoring force that acted on it.



osc_13.png

a 0.35 m s^{-1}

b maximum kinetic energy $= \frac{1}{2} mv^2 = \frac{1}{2} \times 2.0 \times (0.35)^2 = 0.12$ J

c maximum potential energy = maximum kinetic energy = 0.12 J

d maximum acceleration 0.17 m s^{-2}, (from the maximum gradient)

e maximum restoring force $= ma = 2.0 \times 0.17 = 0.34$ N

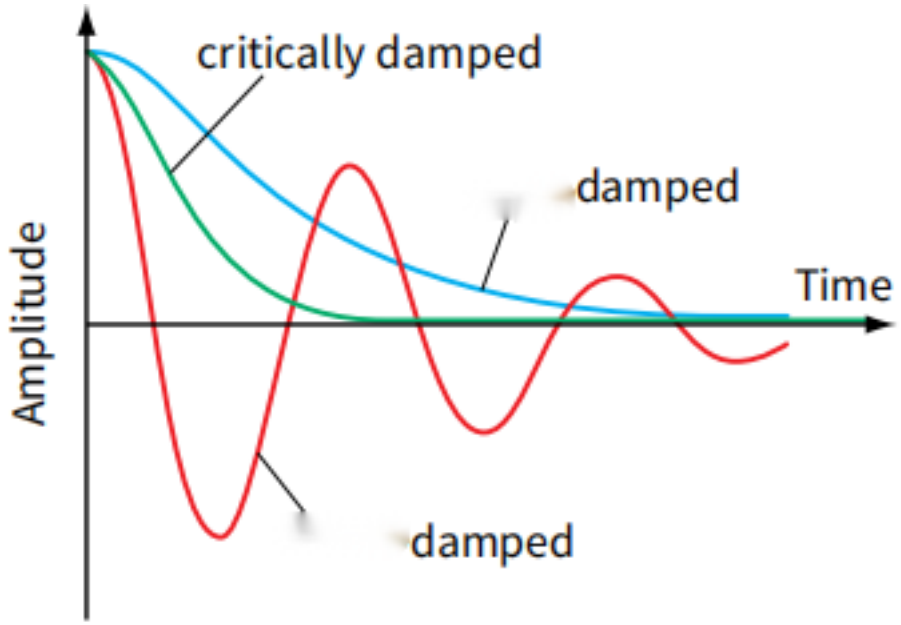


Figure 19.35 Critical damping is just enough to ensure that a damped system returns to equilibrium without oscillating.

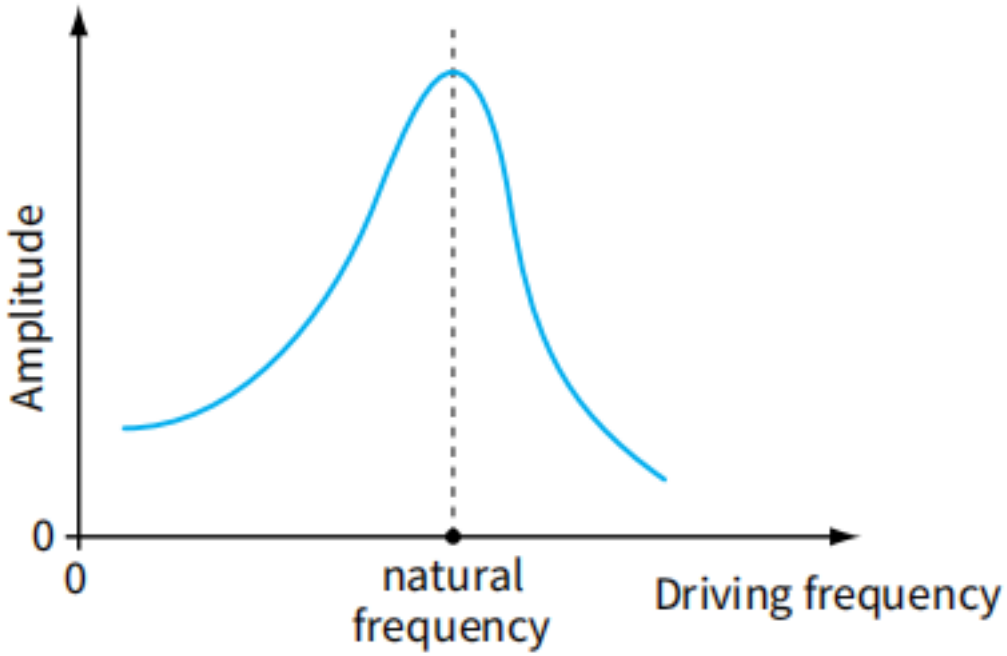
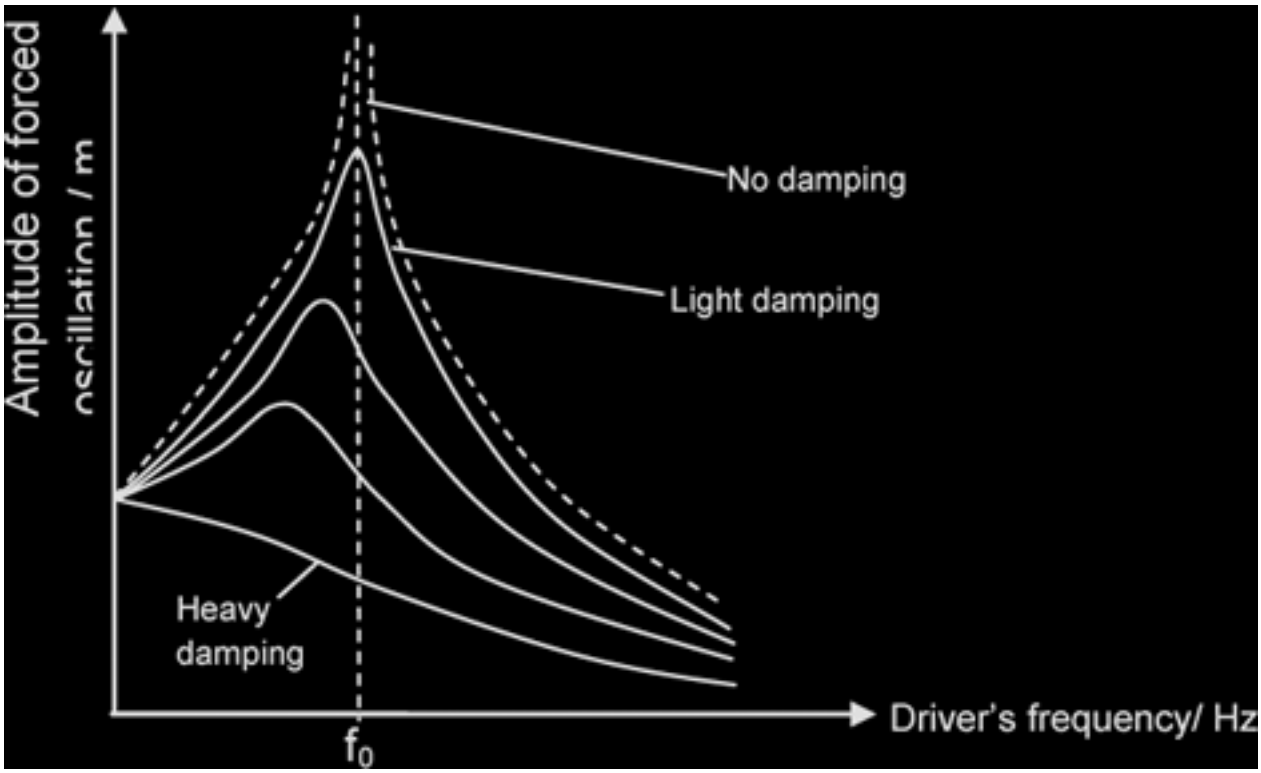
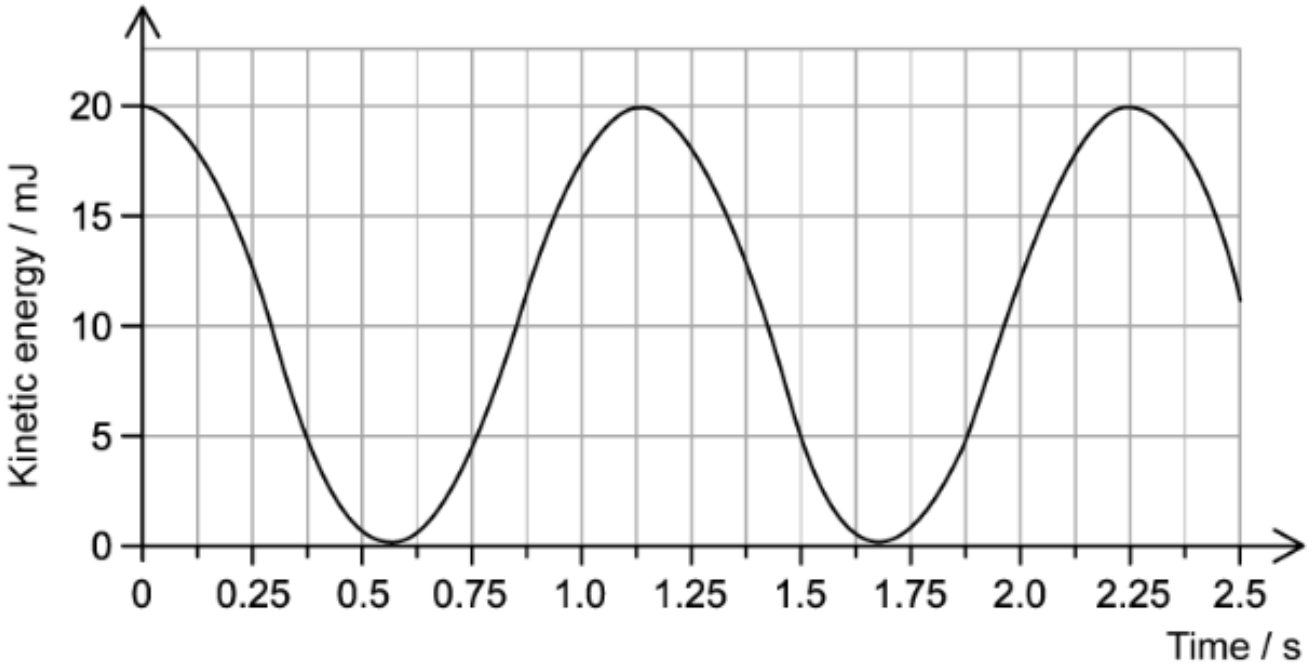


Figure 19.32 Maximum amplitude is achieved when the driving frequency matches the natural frequency of oscillation.

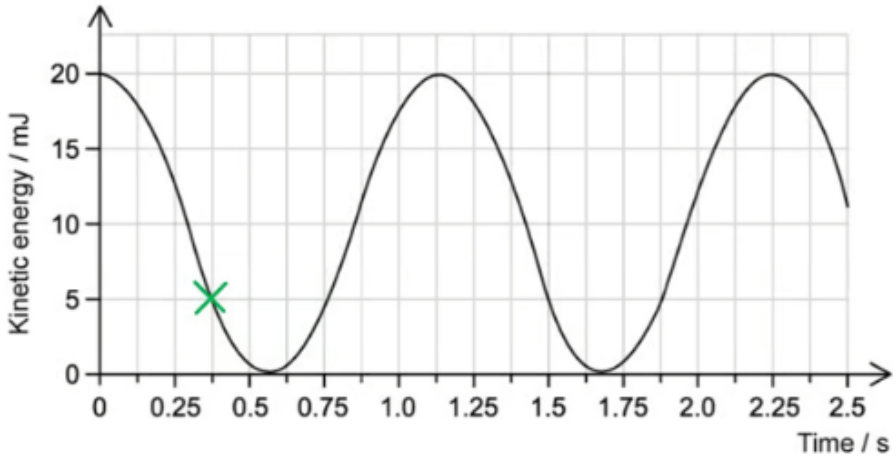

Amplitude of forced oscillation / m
No damping
Light damping
Heavy damping
Driver's frequency/ Hz
f_0

osc_18.png


Kinetic energy / mJ
20
15
10
5
0
0
0.25
0.5
0.75
1.0
1.25
1.5
1.75
2.0
2.25
2.5
Time / s

osc_20.png

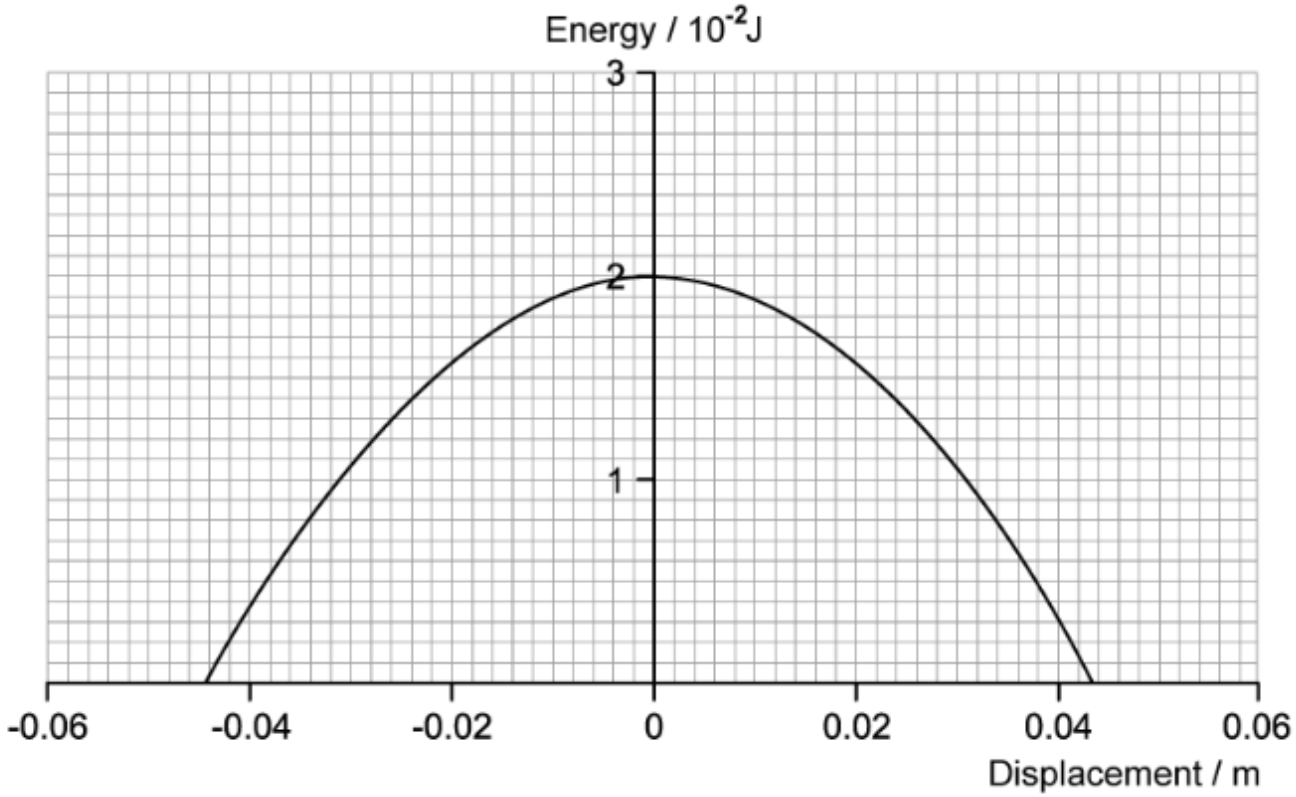
- Kinetic energy is proportional to velocity^2
- If the velocity is halved, then the kinetic energy is a quarter of the initial $\left(\frac{1}{2^2}\right)$
- Any point on the graph at 5 mJ marked; [1 mark]



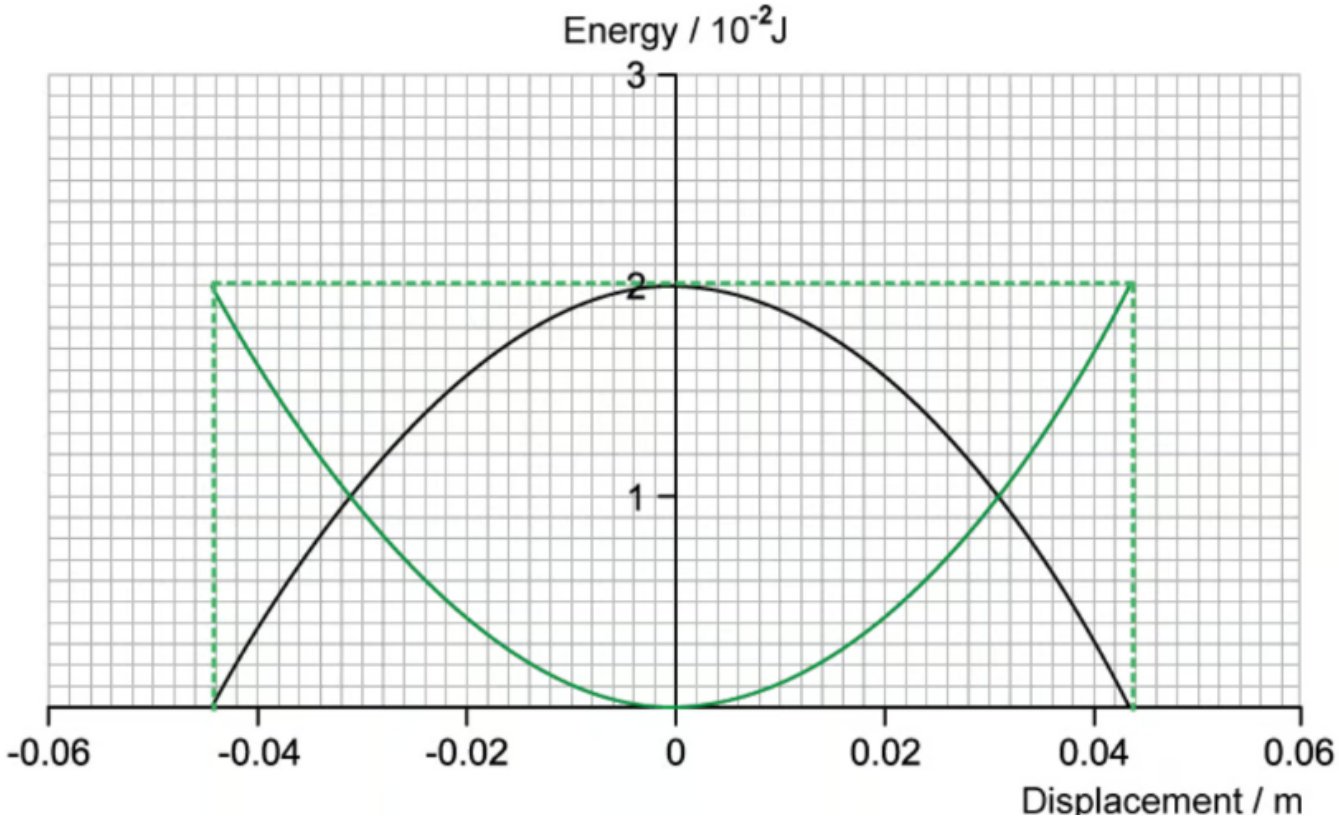
osc_21.png

- From the graph, the time period is 2.25 s
- Acceleration of free fall, g = 9.81 m s^{-2} (from the data page)
- Period, $T = 2\pi\sqrt{\frac{l}{g}}$
- Length, $l = \frac{T^2 g}{4\pi^2} = \frac{(2.25)^2 \times 9.81}{4\pi^2}$ [1 mark]
- l = 1.26 m [1 mark]

Fig. 1.2 shows how the kinetic energy of the pendulum varies with displacement.



osc_24.png



osc_26.png

Calculate the maximum velocity:

- $v = \pm\omega\sqrt{x_0^2 - x^2}$ (from the formula page)
  - At v_{max}, $x = 0$ m
- $v_{max} = \omega x_0 = \frac{2\pi x_0}{T}$
- $v_{max} = \frac{2\pi \times 0.044}{2.25} = 0.12\ \text{m s}^{-1}$ [1 mark]

Calculate mass from equation for kinetic energy:

- $E_{k\ max} = \frac{1}{2} m v_{max}^2$
- $m = \frac{2E_{k\ max}}{v_{max}^2} = \frac{2 \times (2 \times 10^{-2})}{0.12^2}$
- m = 2.78 kg [1 mark]

- Acceleration, $a = -\omega^2 x$ (from the formula page)
  - Maximum acceleration occurs at maximum displacement
- $a_{max} = -\omega^2 x_0 = -\left(\frac{2\pi}{T}\right)^2 x_0$
- $a_{max} = -\left(\frac{2\pi}{2.25}\right)^2 \times 0.044 = -0.343 \text{ m s}^{-2}$ [1 mark]

Calculate the magnitude of the maximum force upon the pendulum:

- $F_{max} = ma_{max}$
- $F_{max} = 2.78 \times 0.343 = 0.95 \text{ N}$ [1 mark]

A mass of 0.42 kg is attached to a spring and the system is made to oscillate with simple harmonic motion (SHM) on a horizontal, frictionless surface. The mass passes through the equilibrium position 200 times per minute.

The kinetic energy of the mass as it passes through the equilibrium position is 500 mJ. There are two points where the restoring force acting on the mass is at its maximum.

Show that the distance between these points is approximately 29 cm.

osc_31.png

Determine the time period of the mass:

- The mass passes through the equilibrium position twice in one time period
  - There are 100 complete oscillations per minute
- $T = \frac{1}{f}$ where $f = \frac{100}{60}$ s^{-1}
- $T = \frac{60}{100} = 0.6$ s [1 mark]

Determine the angular velocity of the mass:

- $\omega = \frac{2\pi}{T} = \frac{2\pi}{0.6} = 10.472$ rad s^{-1} [1 mark]

Kinetic energy (and velocity) is maximum when displacement is zero:

- $E_k = \frac{1}{2}mv^2$
- $v_{max} = \omega x_0$ where x_0 is amplitude
- $E_k = \frac{1}{2}m\omega^2 {x_0}^2$
- $0.5 = \frac{1}{2} \times 0.42 \times (10.472)^2 \times {x_0}^2$

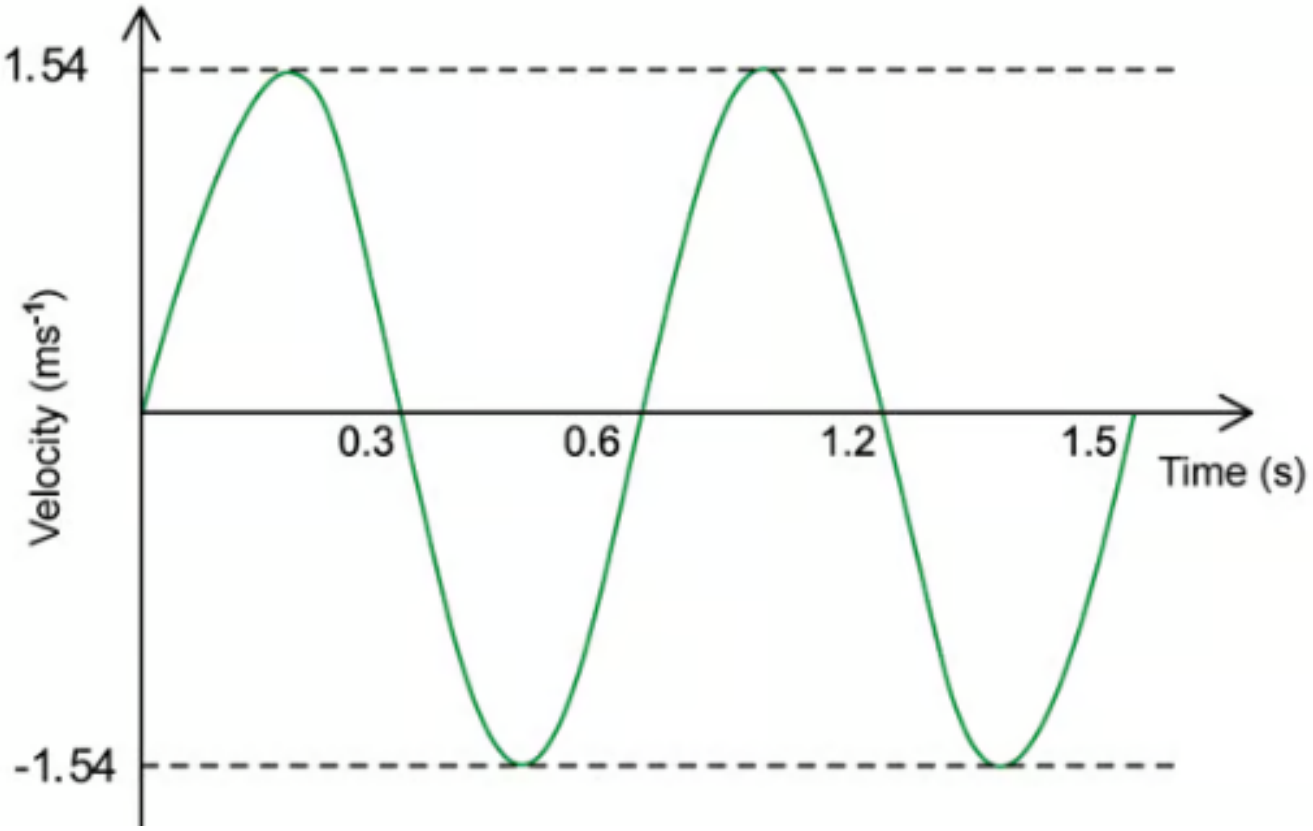
Rearrange for amplitude x_0 and substitute in the values:

- Amplitude, $x_0 = \sqrt{\frac{2 \times 0.5}{0.42 \times (10.472)^2}}$ = 0.147 m = 14.7 cm [1 mark]

Calculate the distance:

- Distance between the two maxima is twice the amplitude
- d = 14.7 × 2 = 29.3 = 29 cm (2 sf) [1 mark]

osc_33.png



osc_35.png

List the known quantities:

- Kinetic Energy at $x = 0$, $E_k = 500$ mJ = 0.5 J
- Mass, $m = 0.42$ kg

Calculate the maximum velocity

- $E_k = \frac{1}{2}mv^2$
- Velocity, $v = \sqrt{\frac{2E_k}{m}} = \sqrt{\frac{2 \times 0.5}{0.42}} = 1.54\ \text{m s}^{-1}$

A graph that would score 2 marks should look like:

State formula for speed of a harmonic oscillator and solve for v:

- $v = \pm\omega\sqrt{\left(x_0{}^2 - x^2\right)}$ (from the formula page)

Rearrange for x:

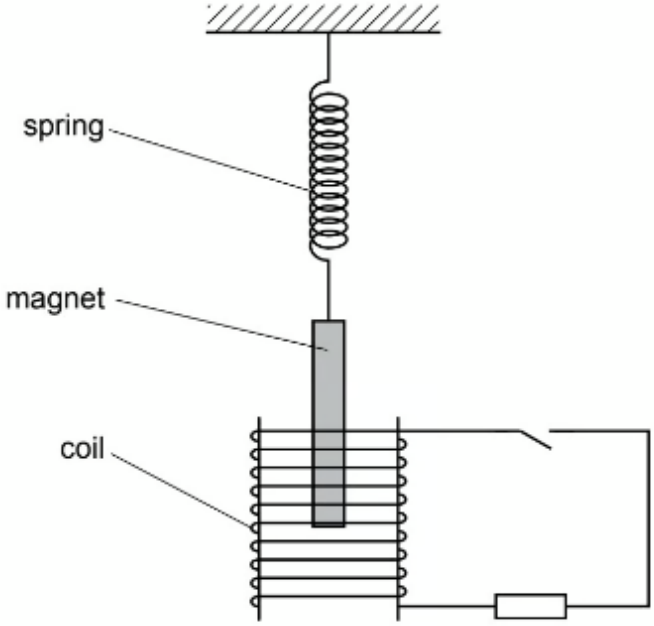
- $\left(\frac{v}{\omega}\right)^2 = x_0{}^2 - x^2$
- $x^2 = x_0{}^2 - \left(\frac{v}{\omega}\right)^2$
- $x = \sqrt{x_0{}^2 - \left(\frac{v^2}{\omega^2}\right)}$ [1 mark]

Substitute the known quantities:

- $x = \sqrt{(0.147)^2 - \left(\frac{(0.8)^2}{(10.5)^2}\right)}$
- $x = 0.126$ m [1 mark]

osc_38.png

A bar magnet of mass 250 g is suspended from the free end of a spring, as illustrated in Fig. 3.1.



osc_39.png

The magnet hangs so that one pole is near the centre of a coil of wire.
The coil is connected in series with a resistor and a switch. The switch is open.

The magnet is displaced vertically and then allowed to oscillate.

At time $t = 0$, the magnet is oscillating freely. At time $t = 6.0$ s, the switch in the circuit is closed.

The variation with time t of the vertical displacement y of the magnet is shown in Fig. 3.2.

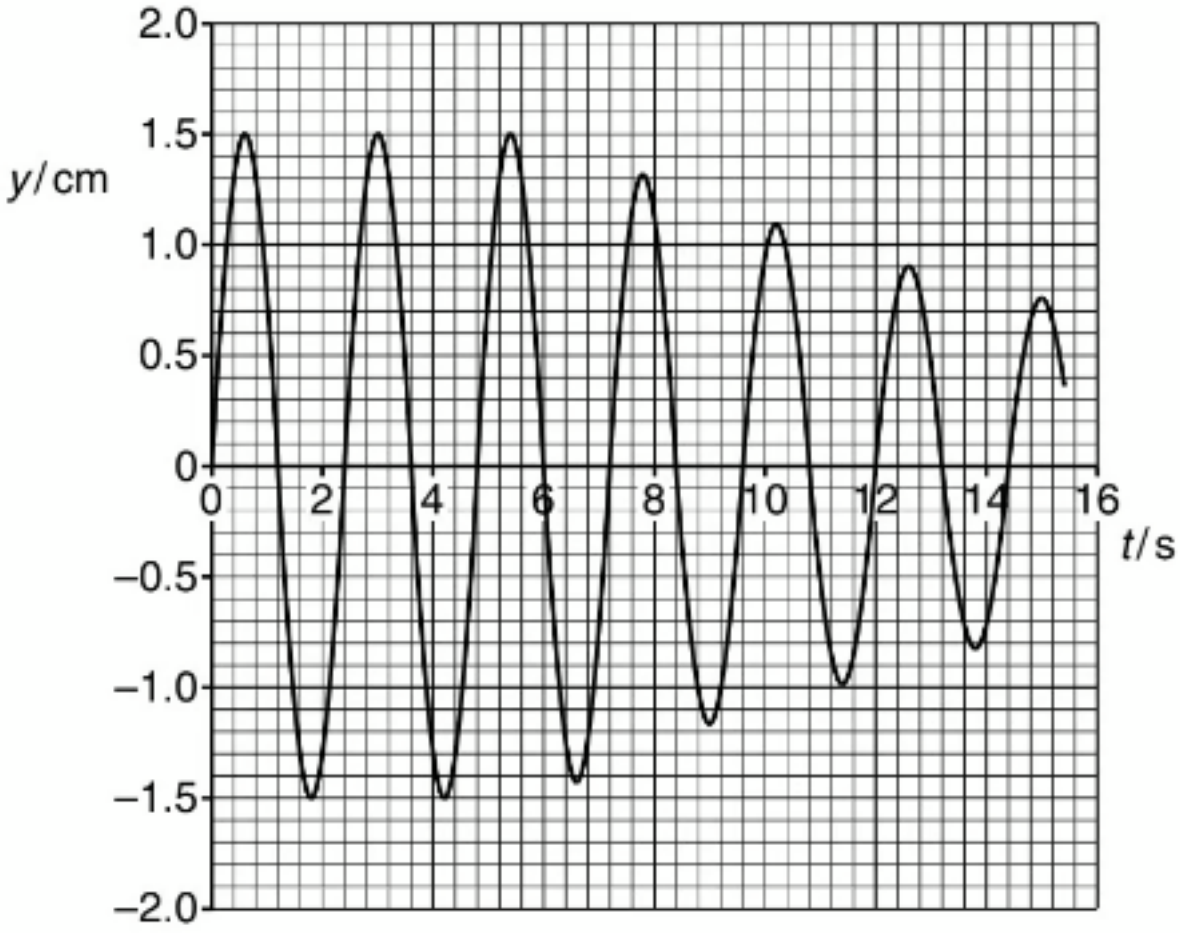


Fig. 3.2

osc_42.png

(ii) the energy of the oscillations during the time interval $t = 0$ to $t = 6.0$ s.

energy = ... J [3]

- Maximum amplitude, $y_0 = 1.5\text{ cm} = 1.5 \times 10^{-2}\text{ cm}$

Write an expression for the total energy in terms of f and y_0:

- $E = \frac{1}{2} m\omega^2 (x_0)^2$
- $\omega = 2\pi f$
- $E = \frac{1}{2} m(2\pi f)^2 (y_0)^2$ [1 mark]

Substitute the known quantities into the expression:

- $E = \frac{1}{2} \times (250 \times 10^{-3}) \times 4\pi^2 \times 0.42^2 \times (1.5 \times 10^{-2})^2$ [1 mark]
- $E = 1.96 \times 10^{-4}\text{ J} = 2.0 \times 10^{-4}\text{ J (2 s.f.)}$ [1 mark]

osc_45.png

(a) State what is meant by resonance.

When an oscillating system is subjected to an external driving force with a frequency equal to the natural frequency of the system, the system will oscillate with maximum amplitude. [2]

(b) Fig. 4.1 shows a heavy pendulum and a light pendulum, both suspended from the same piece of string. This string is secured at each end to fixed points.

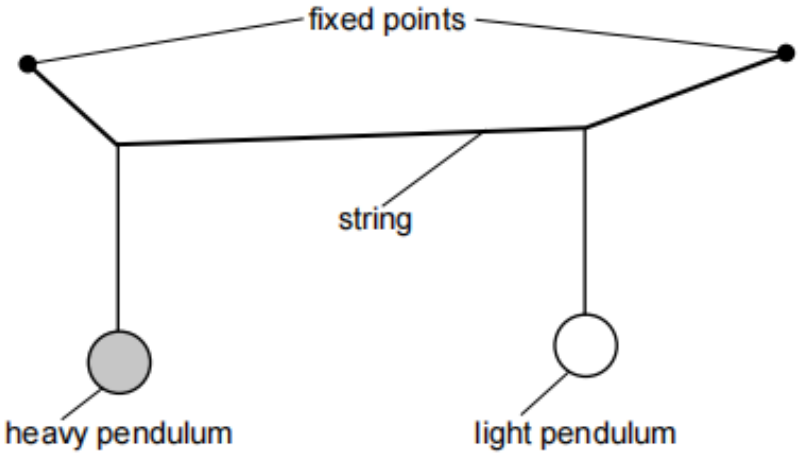


Fig. 4.1

osc_48.png

Both pendulums have the same natural frequency.
The heavy pendulum is set oscillating perpendicular to the plane of the diagram. As it oscillates, it causes the light pendulum to oscillate.

Fig. 4.2 shows the variation with time *t* of the displacements of the two pendulums for three oscillations.

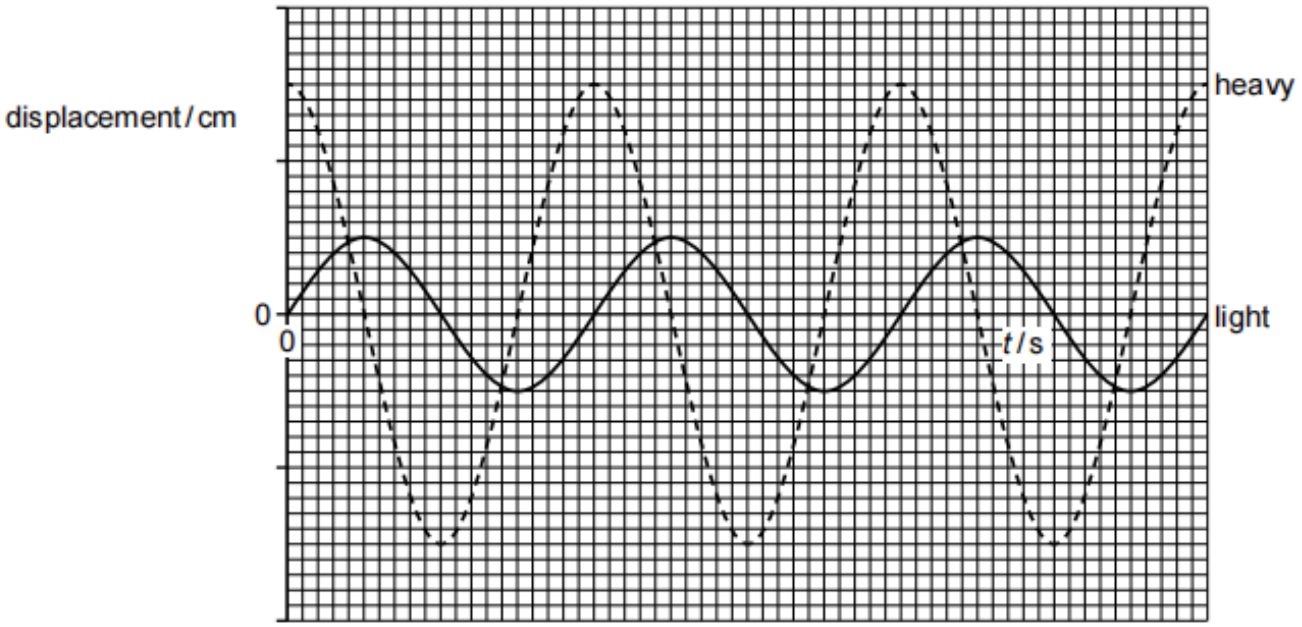


Fig. 4.2

The variation with t of the displacement x of the light pendulum is given by

$$x = 0.25 \sin 5.0\pi t$$

where x is in centimetres and t is in seconds.

(i) Calculate the period T of the oscillations.

$T = 2\pi / \omega = 2\pi / 5.0\pi = 0.40$ s

$T =$ ..... 0.40 ..... s [2]

(ii) On Fig. 4.2, label both of the axes with the correct scales. Use the space below for any additional working that you need.

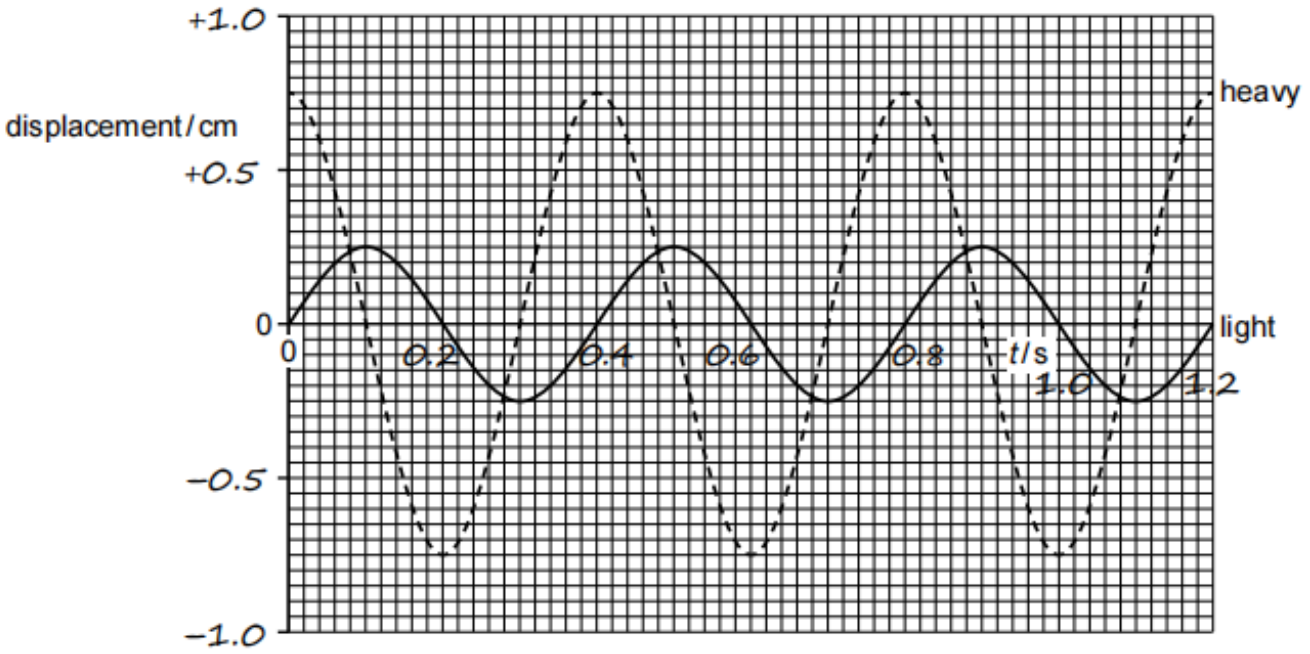


Fig. 4.2

(iii) Determine the magnitude of the phase difference ϕ between the oscillations of the light and heavy pendulums. Give a unit with your answer.

$\emptyset = 2\pi\Delta t / T = 2\pi \times 0.10 / 0.40 = 1.6$ rad

17 Oscillations

17.1 Simple harmonic oscillations

Candidates should be able to:

1 understand and use the terms displacement, amplitude, period, frequency, angular frequency and phase difference in the context of oscillations, and express the period in terms of both frequency and angular frequency

2 understand that simple harmonic motion occurs when acceleration is proportional to displacement from a fixed point and in the opposite direction

3 use $a = -\omega^2 x$ and recall and use, as a solution to this equation, $x = x_0 \sin \omega t$

4 use the equations $v = v_0 \cos \omega t$ and $v = \pm \omega \sqrt{(x_0^2 - x^2)}$

5 analyse and interpret graphical representations of the variations of displacement, velocity and acceleration for simple harmonic motion

17.2 Energy in simple harmonic motion

Candidates should be able to:

1 describe the interchange between kinetic and potential energy during simple harmonic motion

2 recall and use $E = \frac{1}{2} m\omega^2 x_0^2$ for the total energy of a system undergoing simple harmonic motion

17.3 Damped and forced oscillations, resonance

Candidates should be able to:

1 understand that a resistive force acting on an oscillating system causes damping

2 understand and use the terms light, critical and heavy damping and sketch displacement–time graphs illustrating these types of damping

3 understand that resonance involves a maximum amplitude of oscillations and that this occurs when an oscillating system is forced to oscillate at its natural frequency